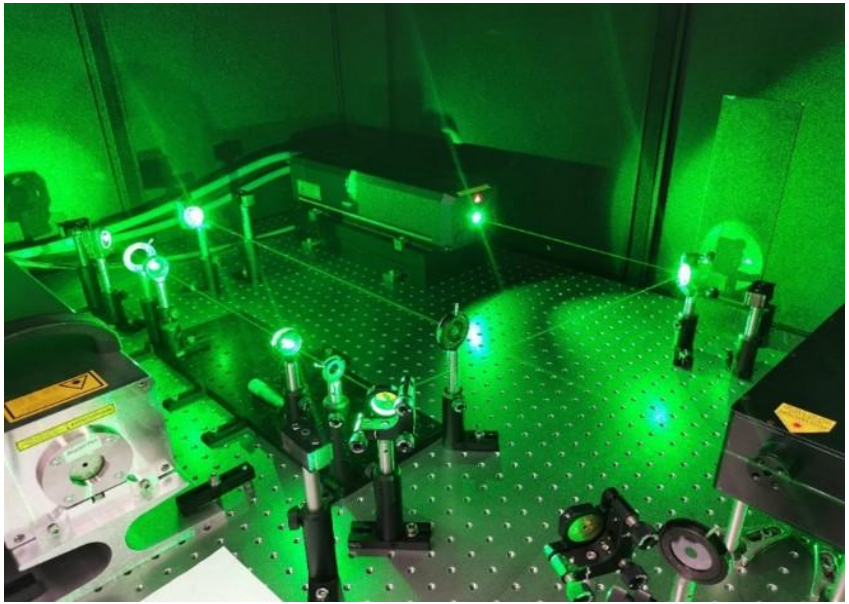
Lasers and Optical Setup for Isobar Suppression in New AMS Facility

Pranav Manoj
FWIR Group
Supervisor: Johannes Lachner
30/08/23

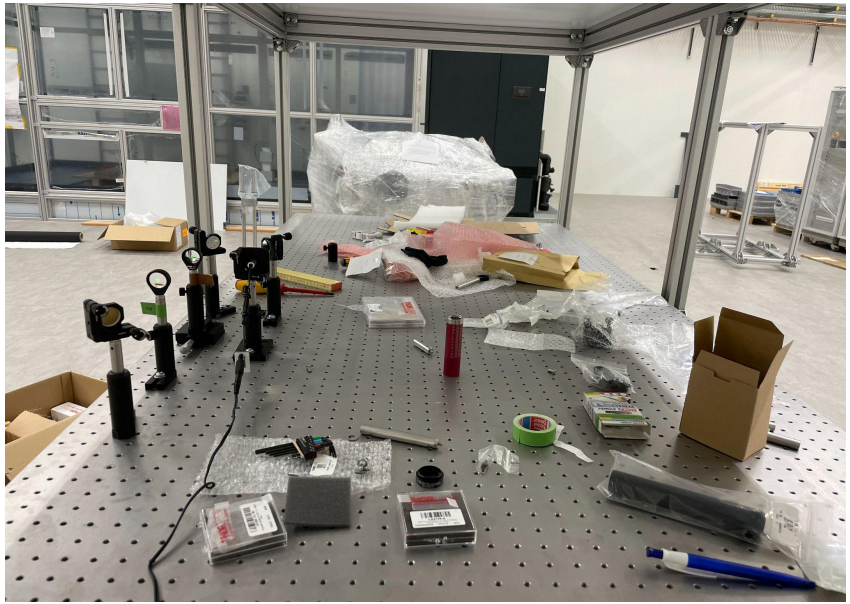
Setting up Lasers for New Accelerator Mass Spectrometry facility

Expectations (end goal):



~ Laser System at Vienna [6]

Reality (so far):



What is Accelerator Mass Spectrometry (AMS)

Measuring/Detecting Radionuclides

- AMS is a technique used to measure long-lived radionuclides with low isotopic abundances [2]



~ DREAMS 6 MV Tandem Accelerator [5]

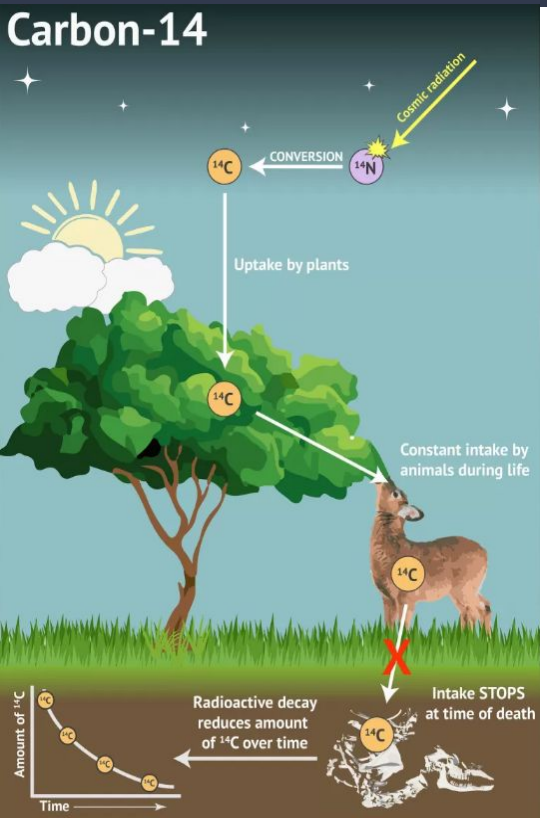
Measuring/Detecting Radionuclides

Counting Atoms of Radioactive Stuff



~ DREAMS 6 MV Tandem Accelerator [5]

Applications of AMS



- ^{14}C (half life ~5700 yrs) used as a dating tool in Archaeology by measuring its content in objects
  - ^{14}C produced by cosmic rays -> biosphere and hydrosphere
- New radionuclides are possible with AMS
- Geology, Archaeology, Astrophysics, etc.

~ Carbon-14 production in the atmosphere and its decay in the biosphere [7]

Two Ways to Detect Radionuclides

Detect radiation emitted in their decay

Pros:

- Much less expensive
- Does not require fancy chemistry (for gamma measurements)

Cons:

- Doesn't work well for longer half lifes and low abundances ($A \sim N/t_{1/2}$)

Directly count atoms with Accelerator Mass Spectrometry

Pros:

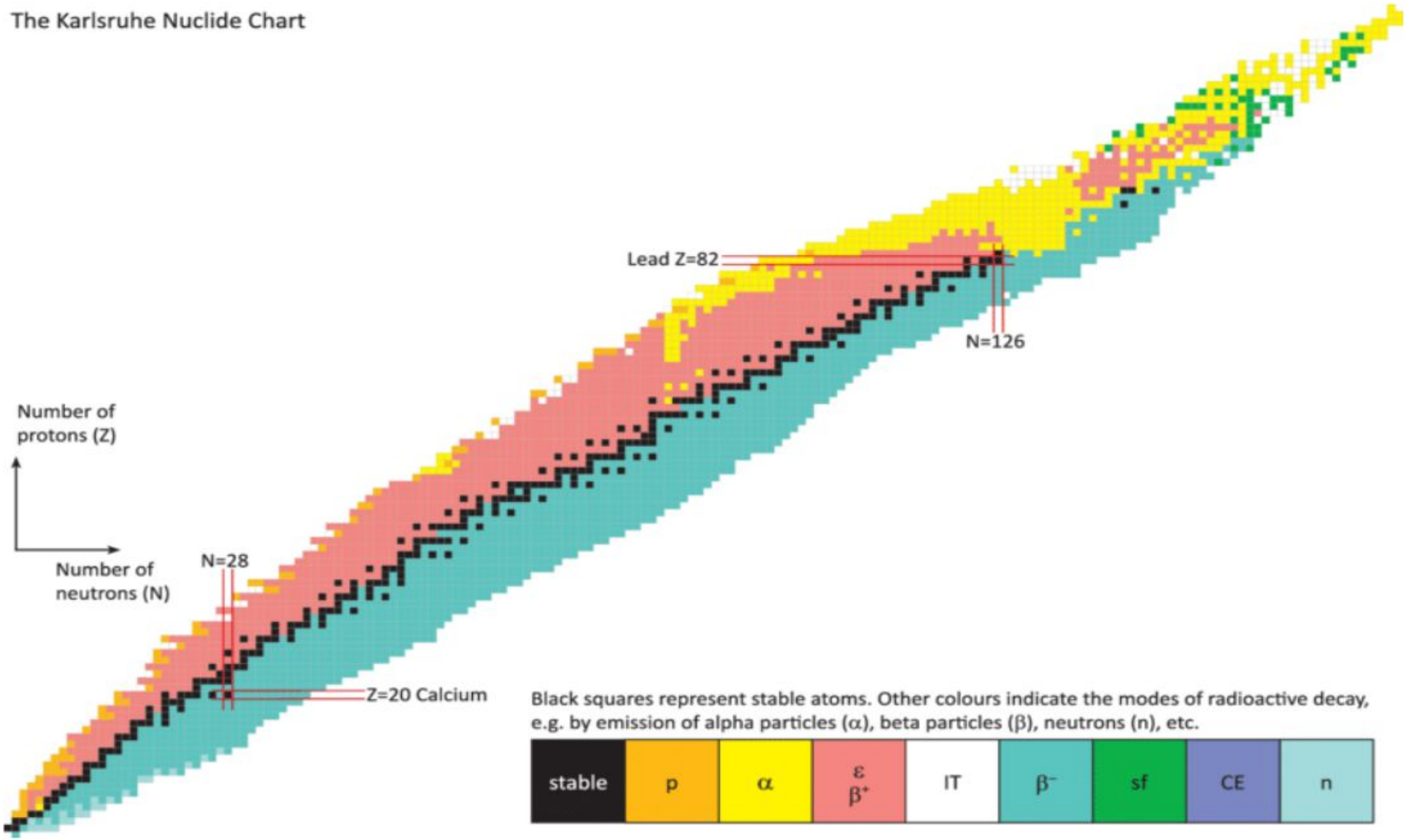
- Half-life of radionuclide is not a concern
- Capable of performing isotope ratio measurements of nuclides at 10^{-12} to 10^{-16} relative to the major stable isotope

Cons:

- Atomic isobars interfere with measurements

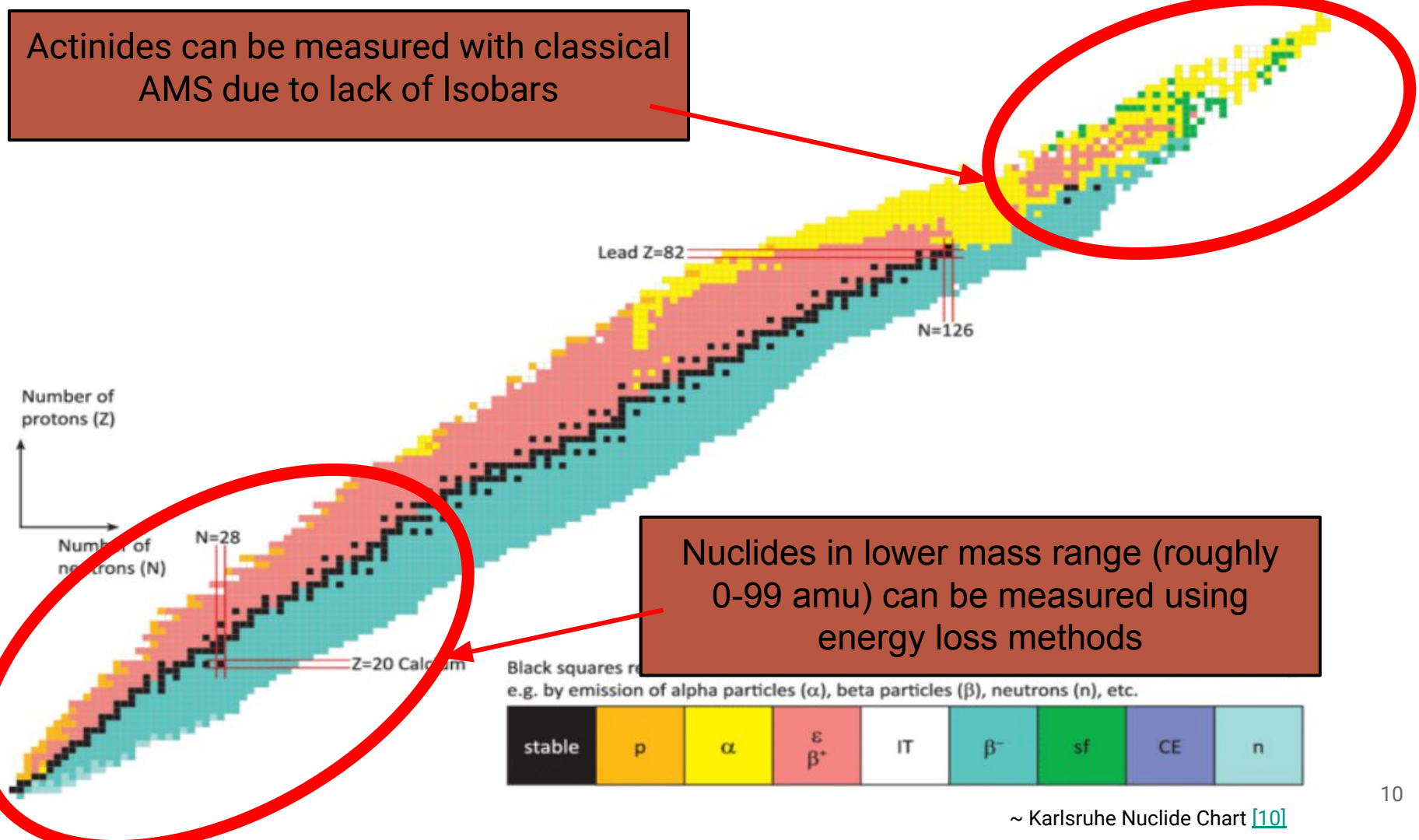
The Isobar Problem

- **Isobars** are atoms (nuclides) of different chemical elements that have the same number of nucleons and thus the same mass number
  - Ex. ^{10}B has the same mass number as ^{10}Be
  - Major challenge for AMS to separate isobars from target



~ Karlsruhe Nuclide Chart [10]

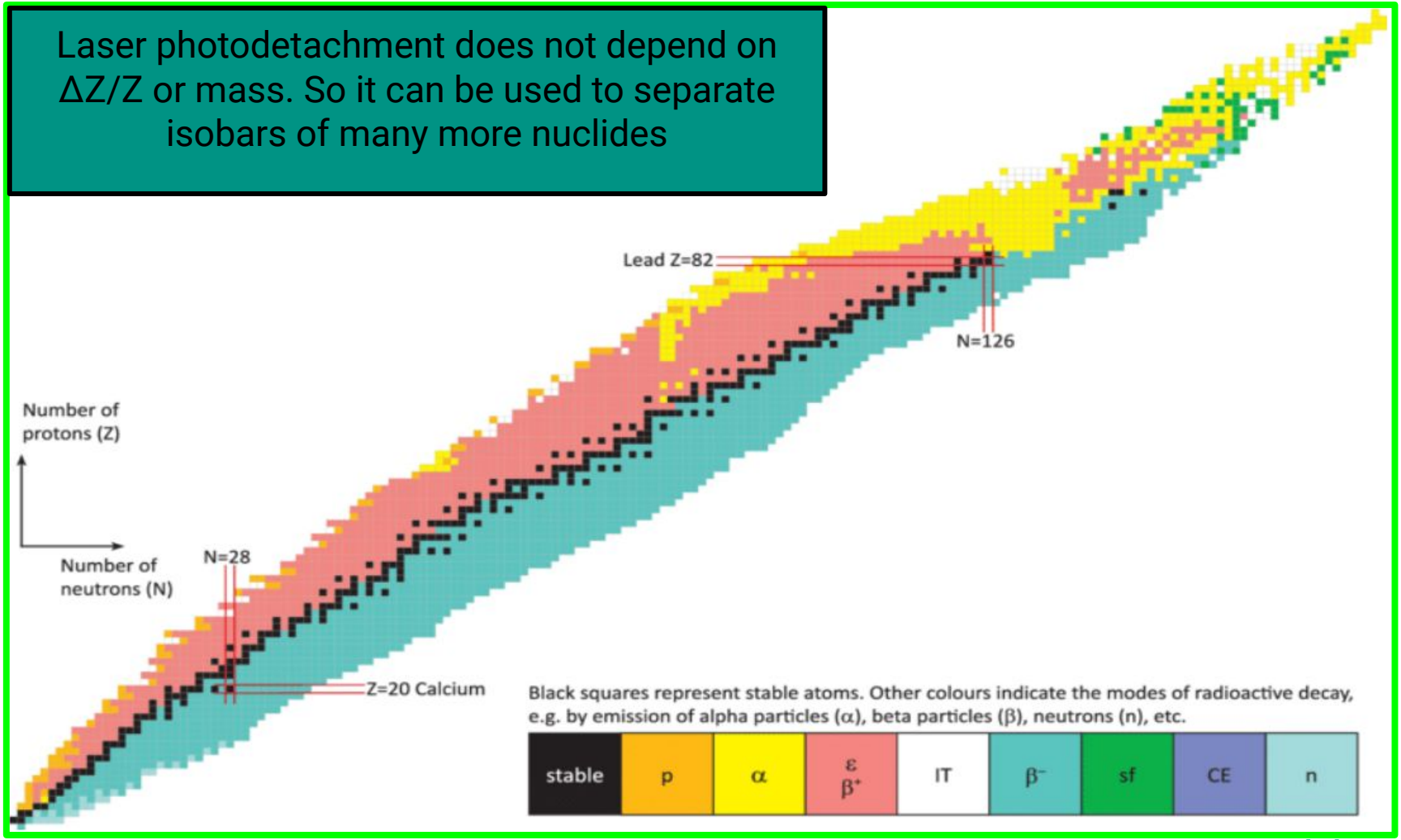
Actinides can be measured with classical AMS due to lack of Isobars



Nuclides in lower mass range (roughly 0-99 amu) can be measured using energy loss methods

~ Karlsruhe Nuclide Chart [10]

Laser photodetachment does not depend on ΔZ/Z or mass. So it can be used to separate isobars of many more nuclides



~ Karlsruhe Nuclide Chart [10]

Laser Photodetachment to Separate Isobars

Laser Photodetachment

- Technique for negative ion beams exploit the differences in electronic structure of the ions [1]
  - require slow negative ion beams
  - similar isobar separation powers for heavy nuclides as for light ones
  - Doesn’t depend on relative differences in atomic numbers.



Laser Photodetachment

- **Electron Affinity -** the amount of energy freed when an electron is added to a neutral atom to form a negatively charged ion
  - Higher EA = atom is more likely to hold on to electrons
- Separate isobars with lower electron affinities using laser photodetachment

N = # of surviving ions with EA less than photon energy

$$N/N_0 = e^{-\sigma\varphi t}$$

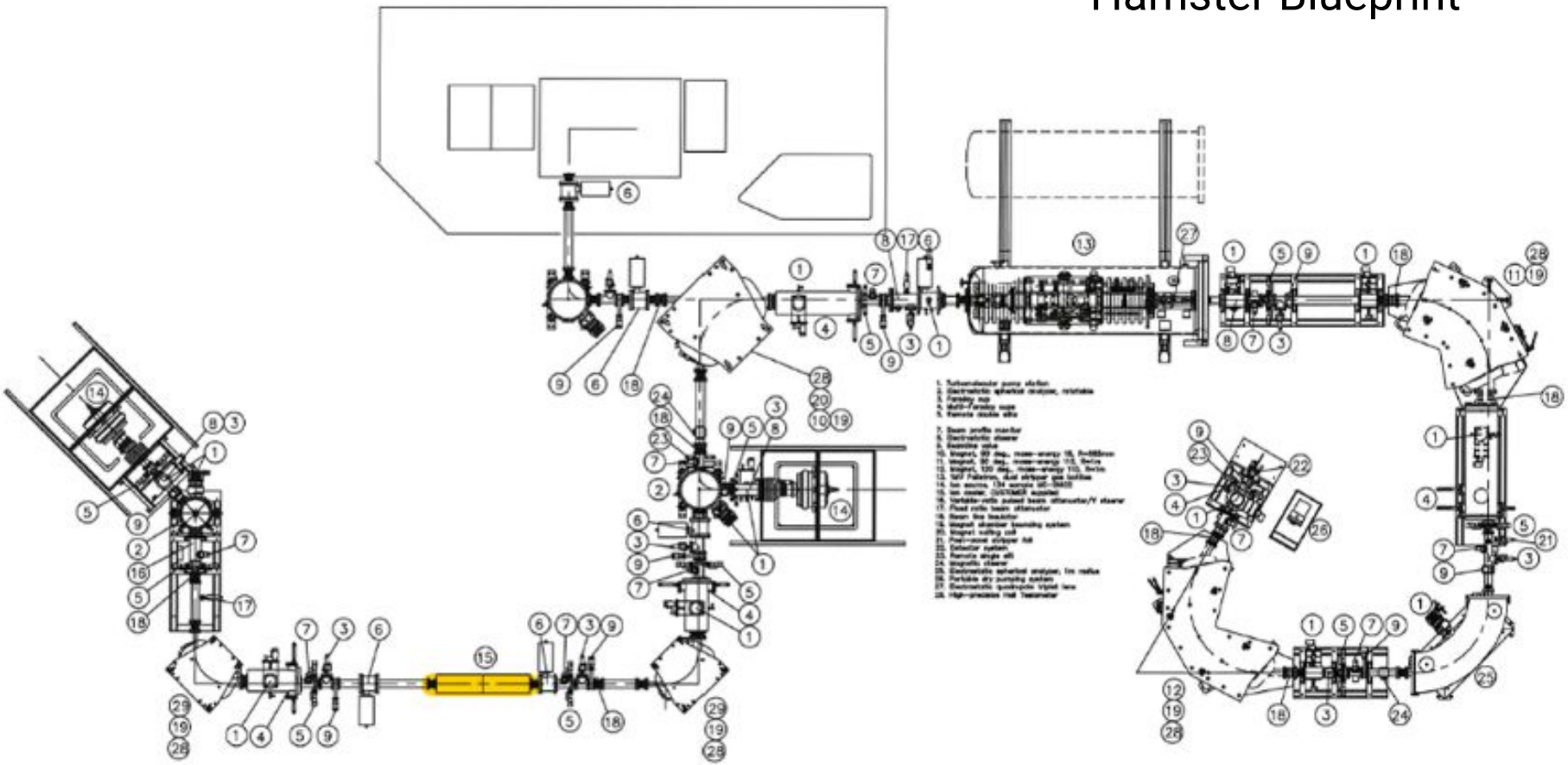
σ = cross section

φ = photon flux

t = interaction time

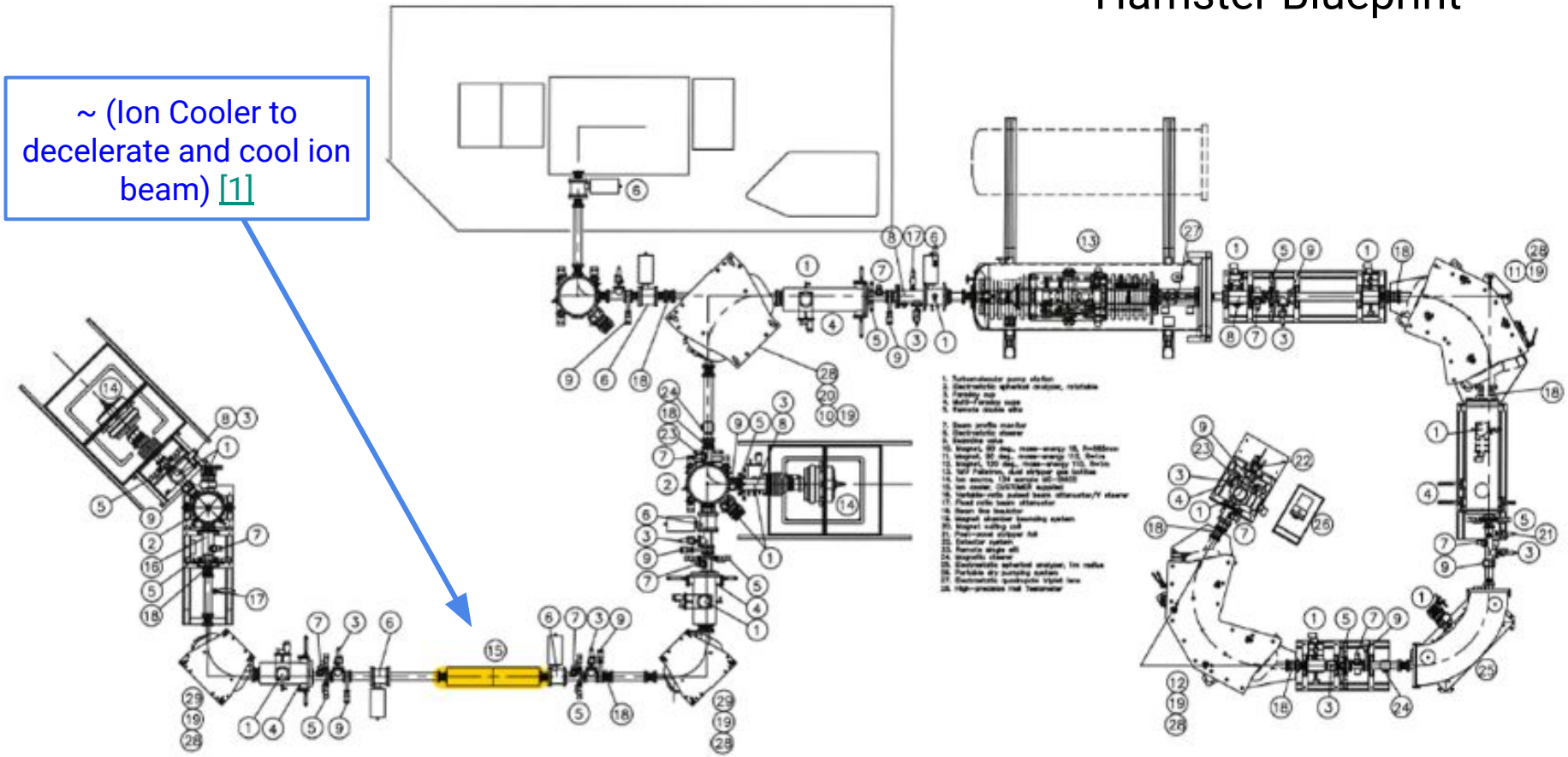
Need to maximize!

Hamster Blueprint

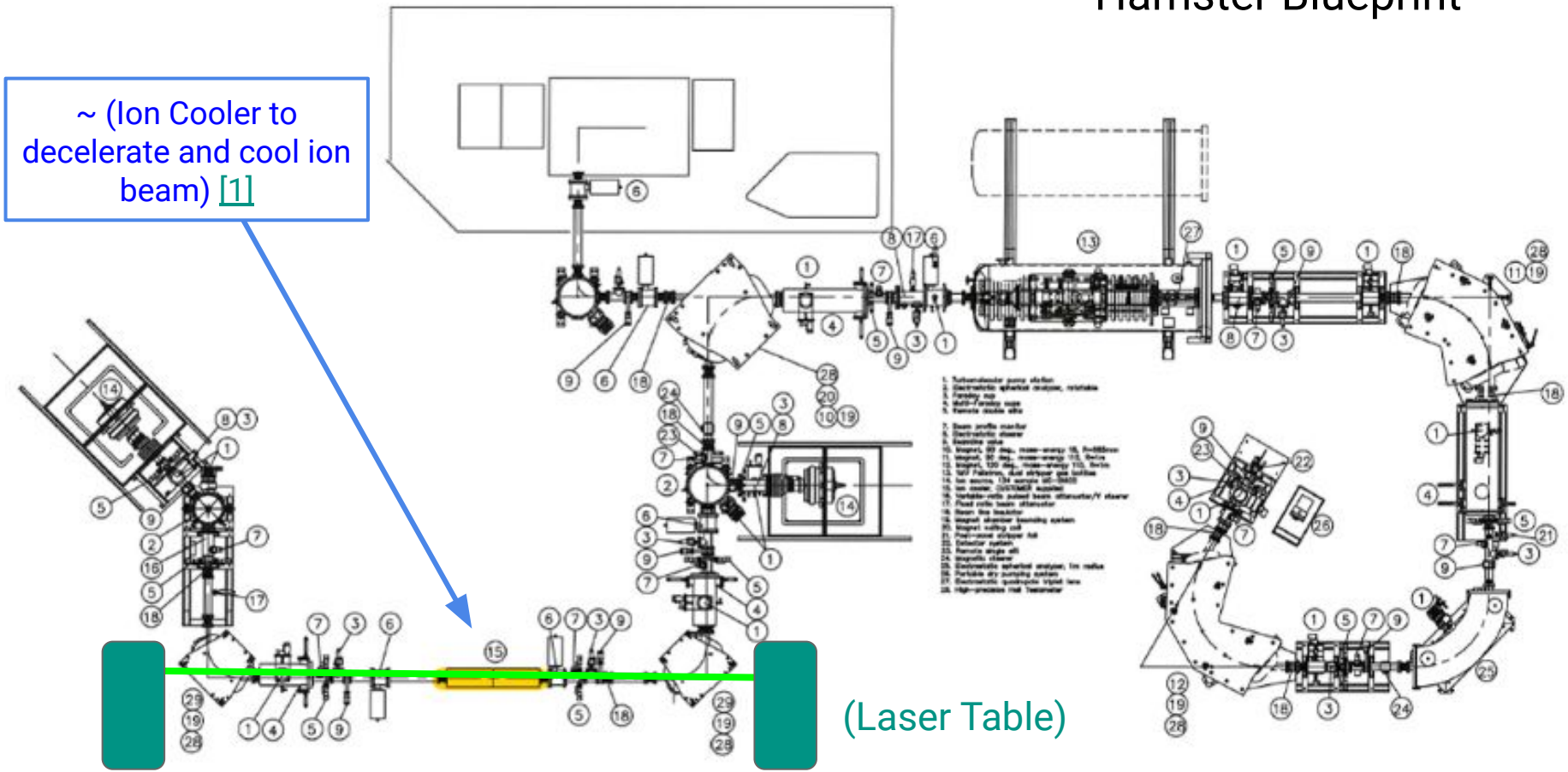


Hamster Blueprint

~ (Ion Cooler to decelerate and cool ion beam) [1]



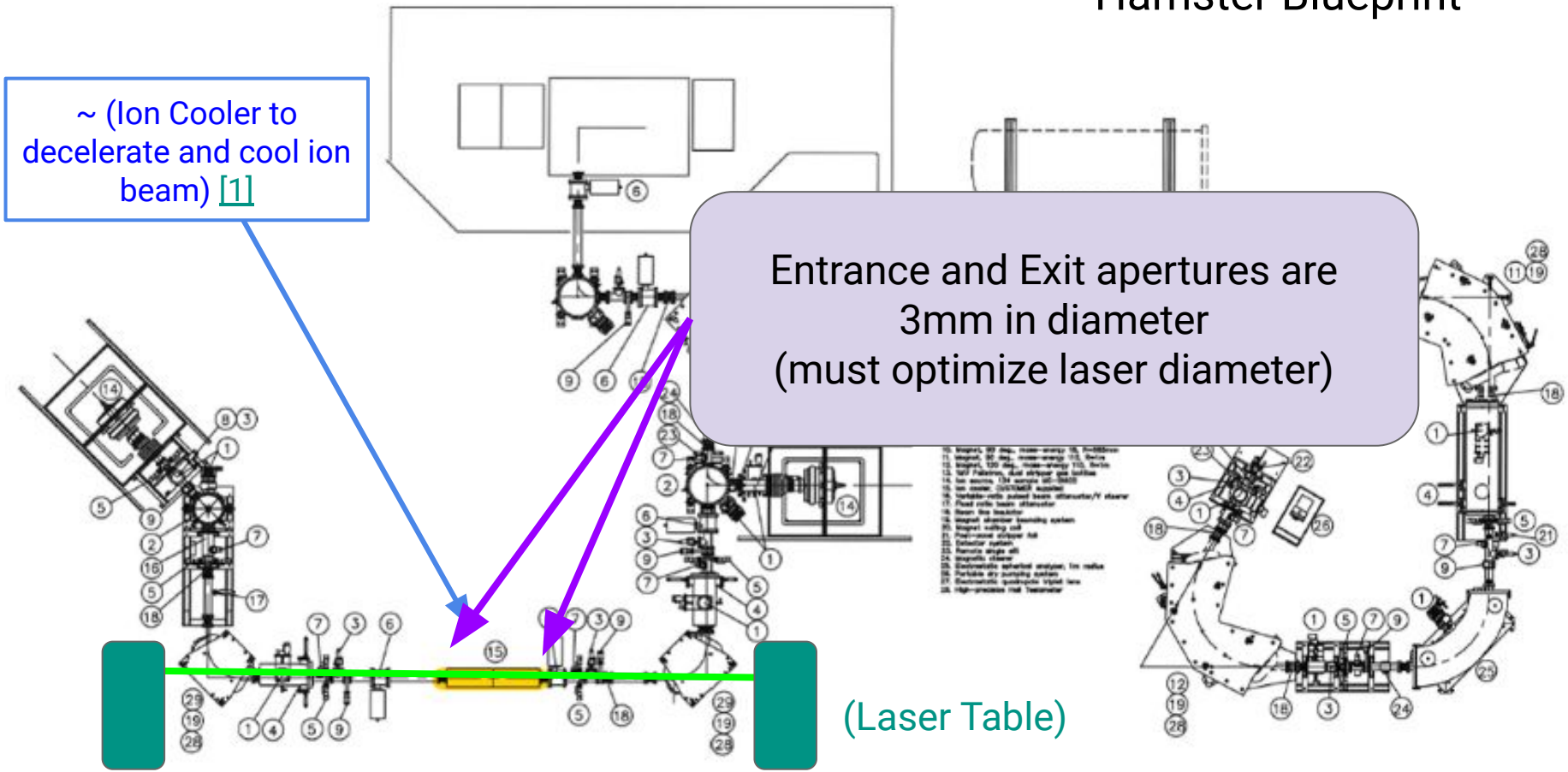
Hamster Blueprint



Hamster Blueprint

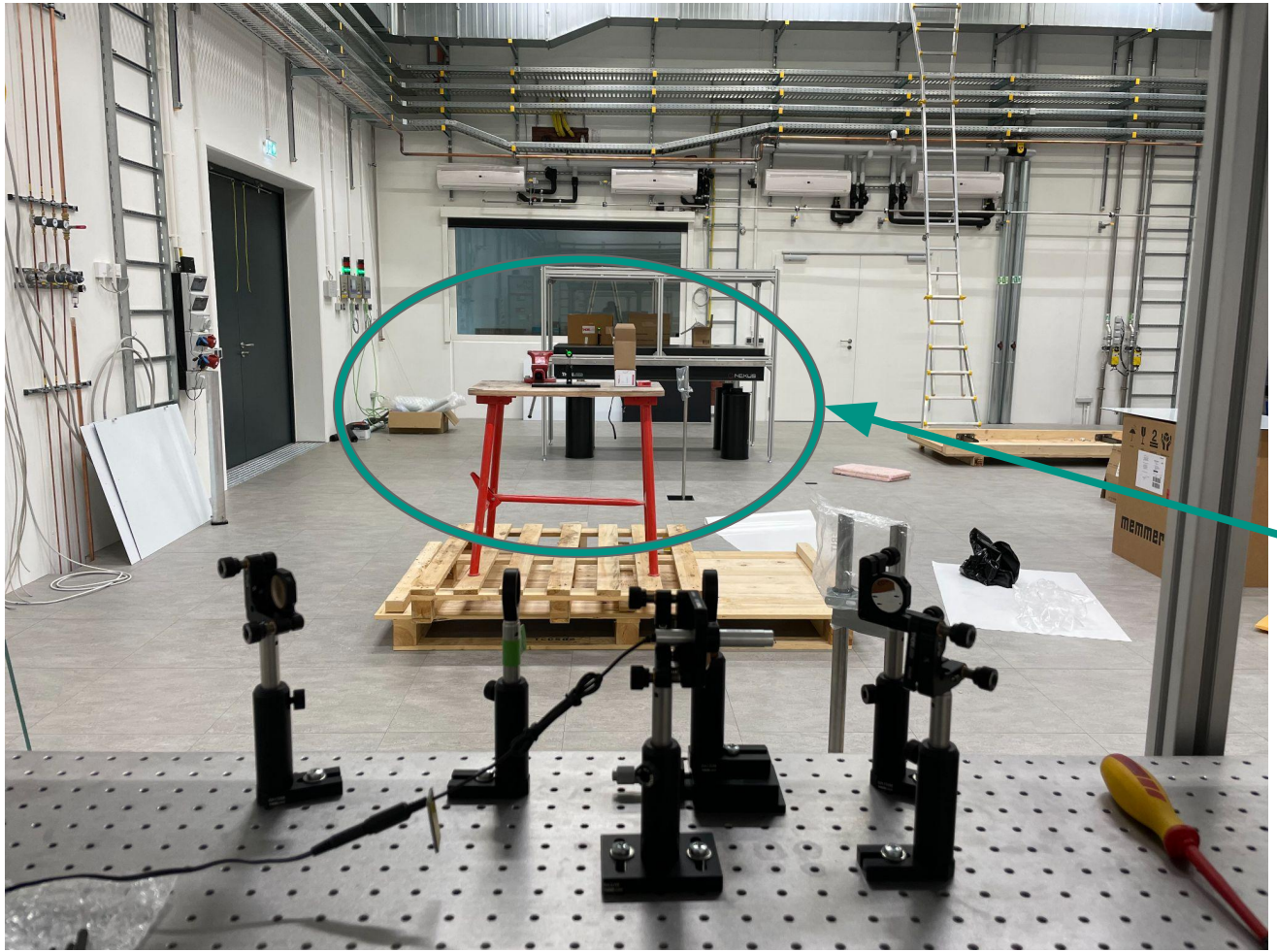
~ (Ion Cooler to decelerate and cool ion beam) [1]

Entrance and Exit apertures are 3mm in diameter
(must optimize laser diameter)



(Laser Table)

Laser Table and Housing



Lasers being Used

- Laser for final setup will be Coherent Verdi C-Series [14]
  - Power: 12 W
  - Wavelength: 532 nm (optical)
  - Beam Diameter: 2.25 mm
  - Beam Divergence: <0.4 mrad

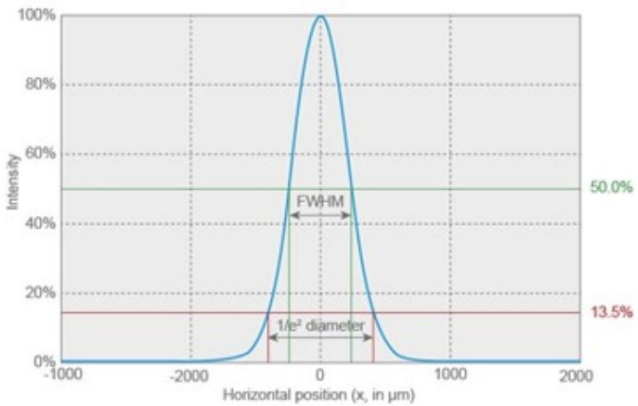


- Laser for testing optics is the ThorLabs compact laser module CPS532-C2 [13]
  - Power: 0.9 mW
  - Wavelength: 532 nm (optical)
  - Beam Diameter: 3.5 mm
  - Beam Divergence: <0.5 mrad

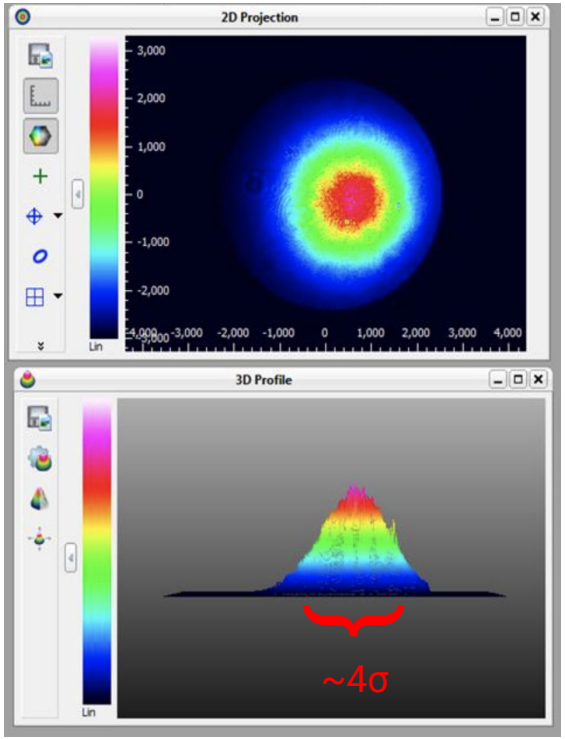


Laser Spot Size/Shape

- Beam Power profile is roughly Gaussian
  - Beam diameter is measured as $1/e^2$ width or about 4σ (corresponds to 95% of beam power)



~ Left: $1/e^2$ beam width [11] , Right: Power profile of test laser [13]



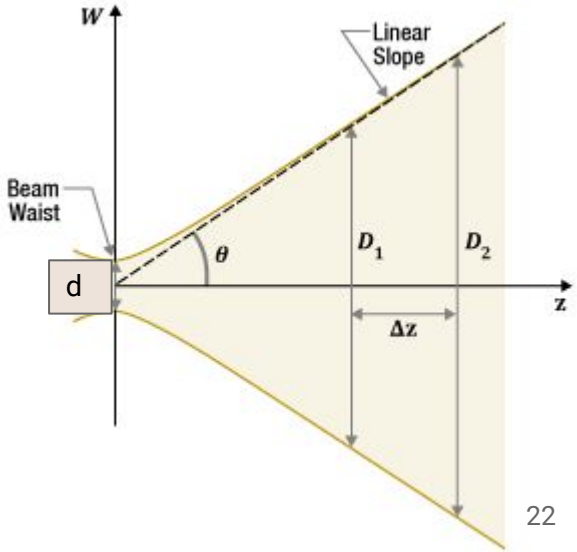
Laser Shape and Divergence

- Beam Divergence causes beam diameter to increase over propagation distance

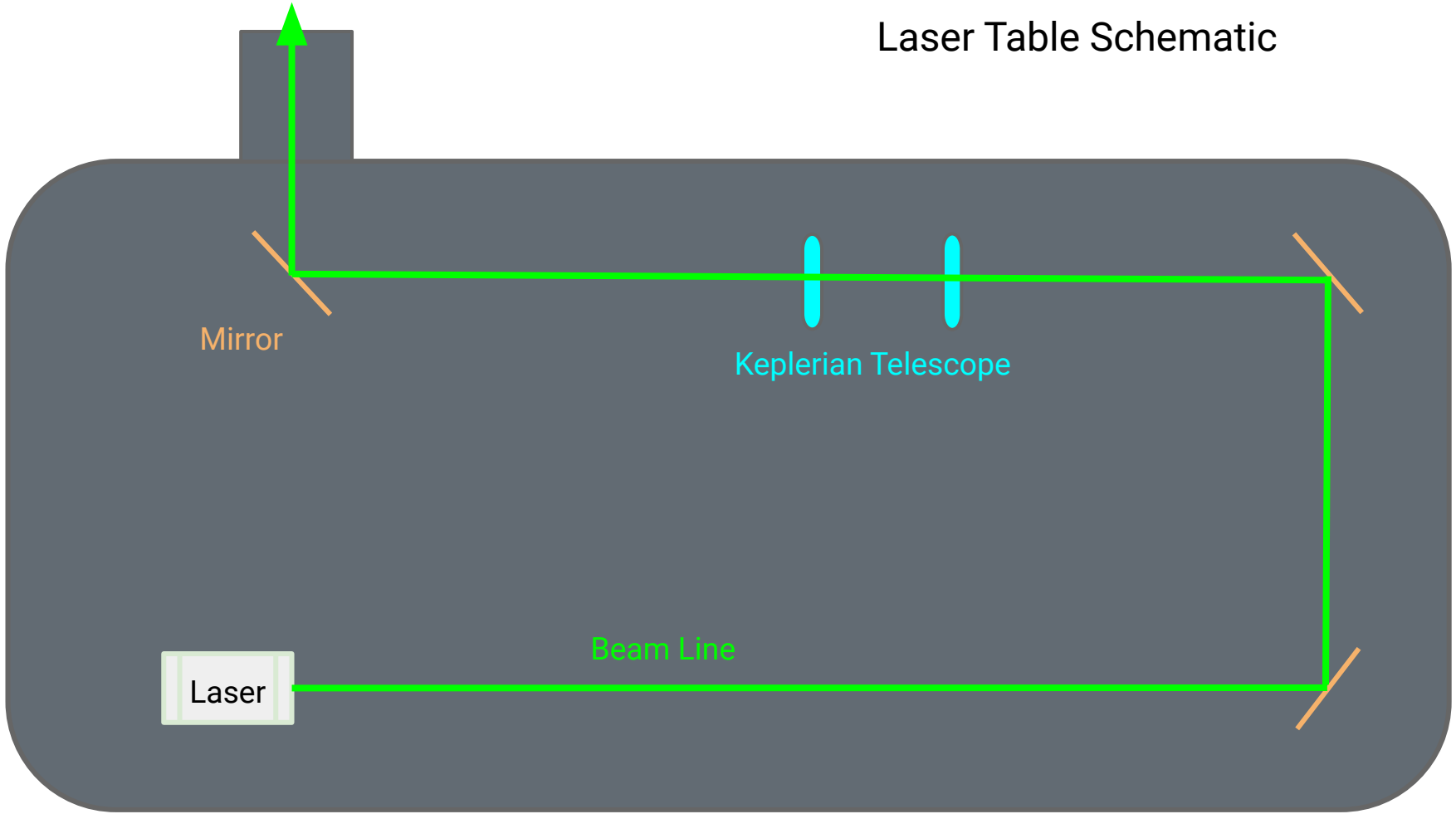
Need to minimize z to get small diameter at ion cooler entrance!

Beam Diameter Equation

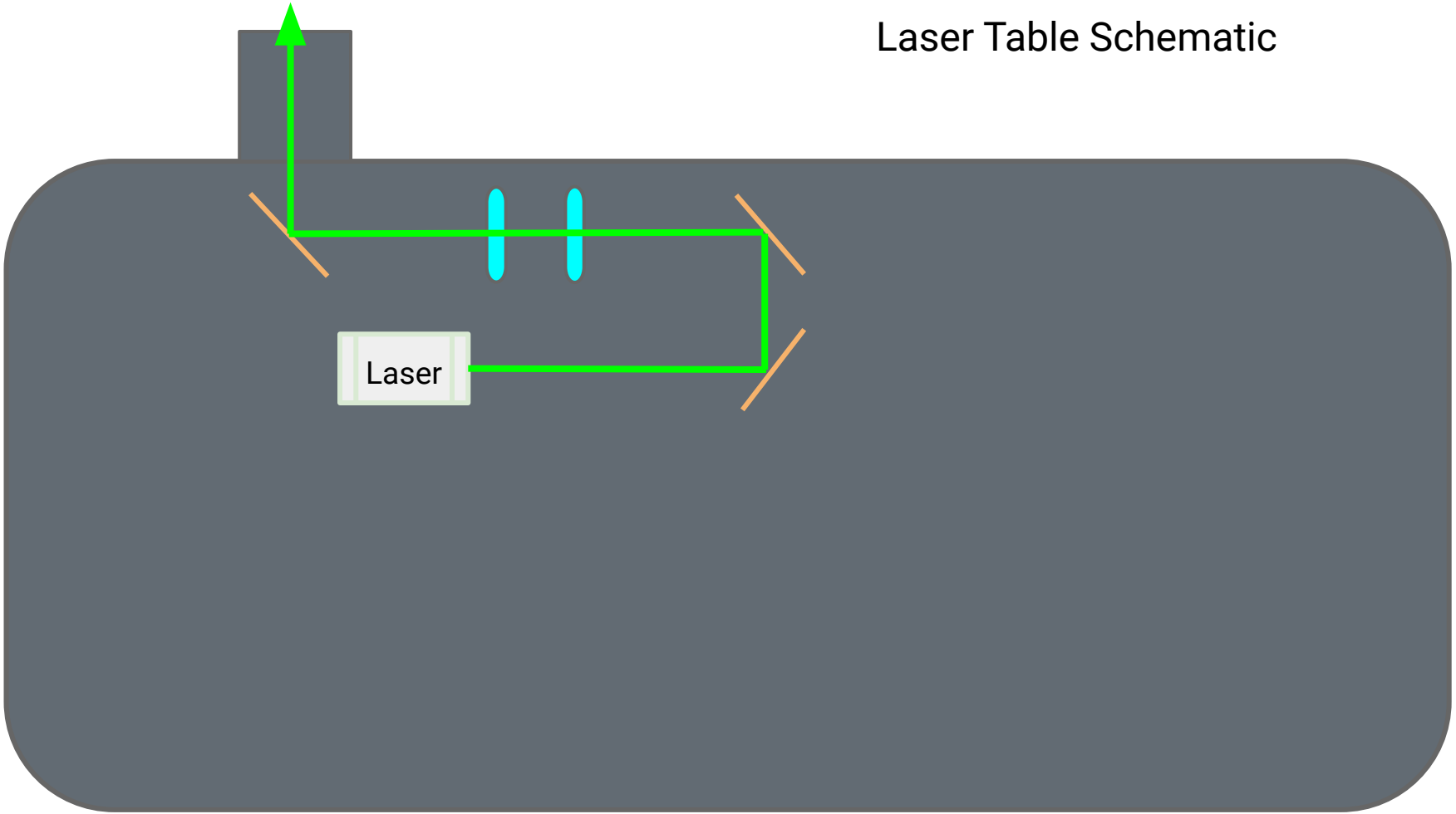
$$D_1 = d + 2*z*\tan(\theta)$$



Laser Table Schematic

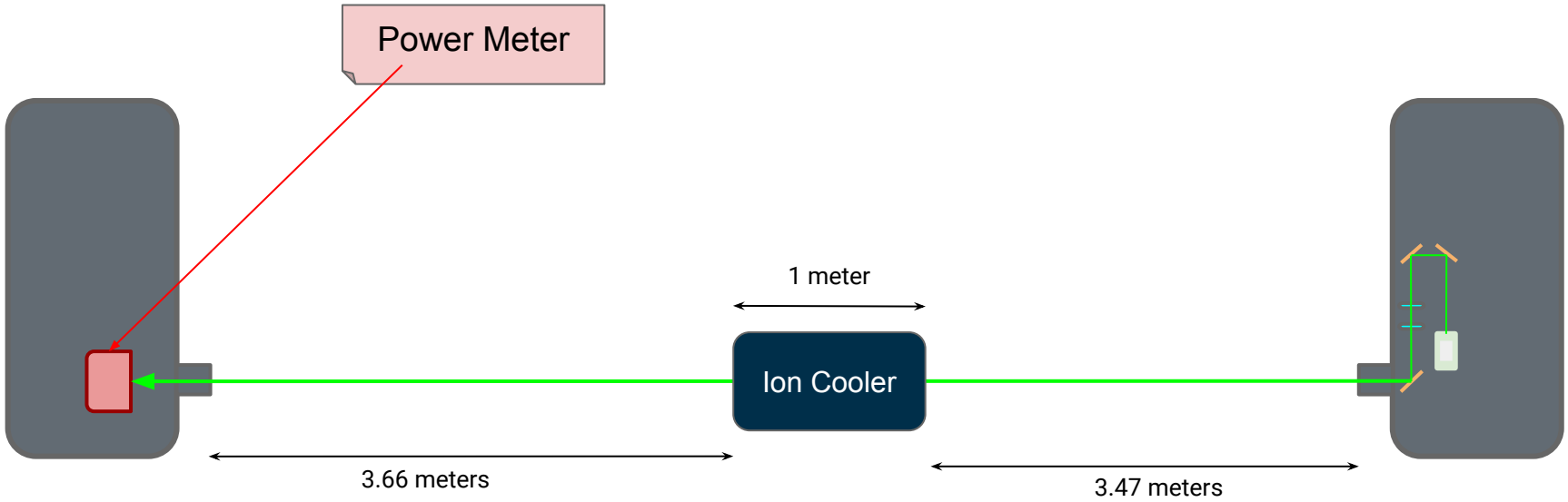


Laser Table Schematic

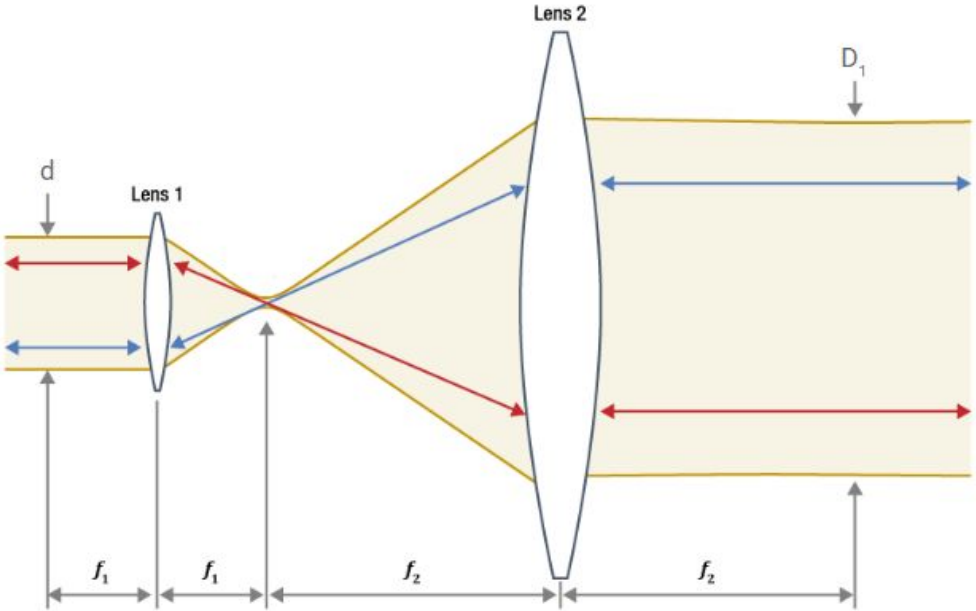




Laser Table Schematic:



Keplerian Telescope



~ Keplerian Beam Expander [12]

- Arrange two convex lenses a distance apart equal to the sum of their focal lengths
- The magnification is given by:

$$m = f_2 / f_1$$

- Keplerian Telescope Effect on Divergence:

$$\Theta_2 = \Theta_1 / m$$

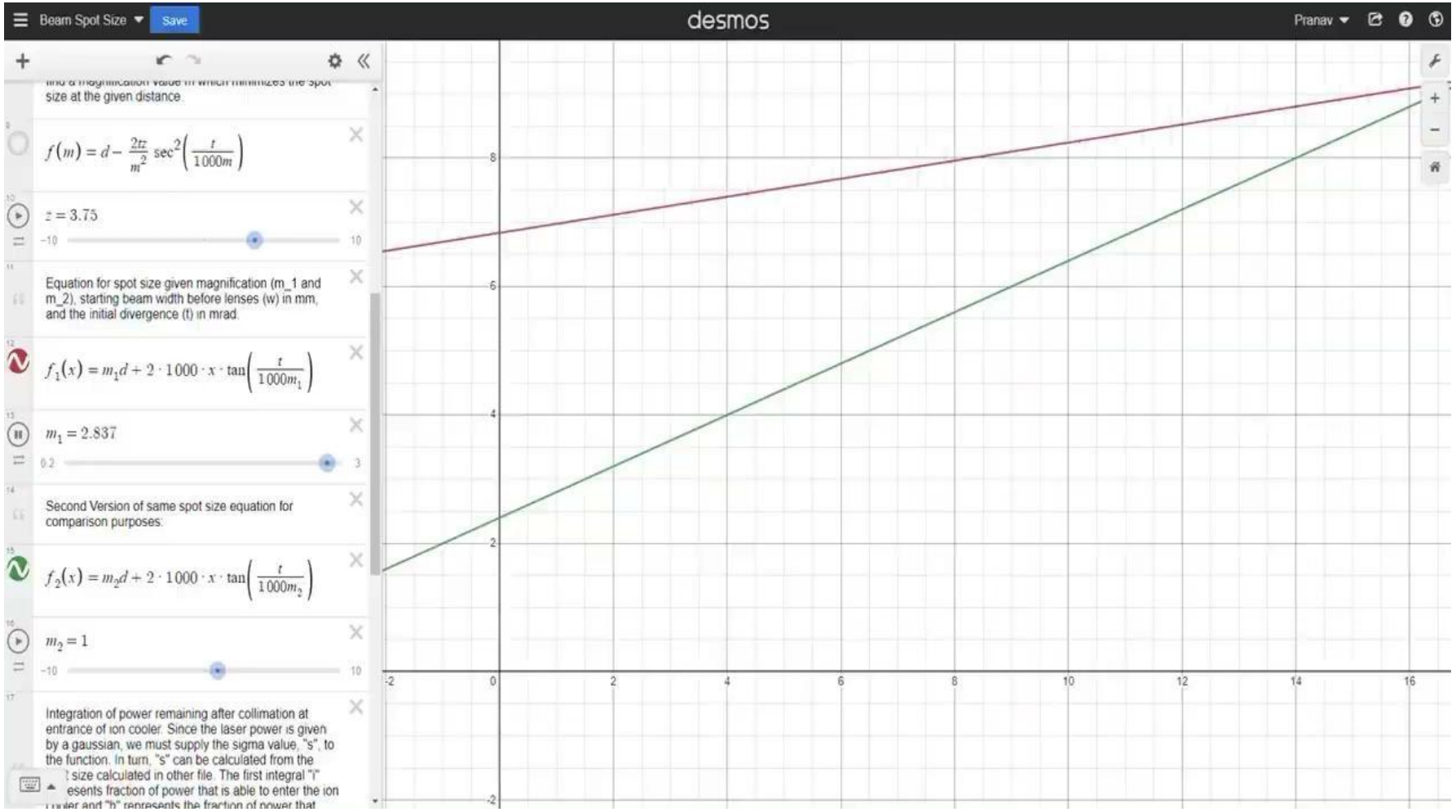
Optimizing Lenses

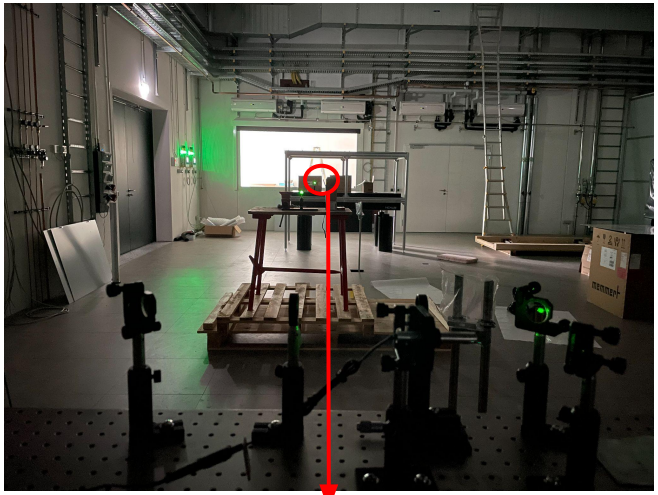
Beam expanders decrease divergence
Beam reducers increase divergence

- With Keplerian telescope, the beam diameter equation becomes:

$$D_1 = d*m + 2*z*\tan(\theta_1/m)$$

- Calculations to find optimal magnification: https://www.desmos.com/calculator/j83oyvsvw6


Beam Spot Size
Save
desmos
Pranav
size at the given distance.
$f(m) = d - \frac{2tz}{m^2}\sec^2\left(\frac{t}{1000m}\right)$
$z = 3.75$
-10
10
Equation for spot size given magnification (m_1 and m_2), starting beam width before lenses (w) in mm, and the initial divergence (t) in mrad.
$f_1(x) = m_1 d + 2 \cdot 1000 \cdot x \cdot \tan\left(\frac{t}{1000m_1}\right)$
$m_1 = 2.837$
0.2
3
Second Version of same spot size equation for comparison purposes:
$f_2(x) = m_2 d + 2 \cdot 1000 \cdot x \cdot \tan\left(\frac{t}{1000m_2}\right)$
$m_2 = 1$
-10
10
Integration of power remaining after collimation at entrance of ion cooler. Since the laser power is given by a gaussian, we must supply the sigma value, "s", to the function. In turn, "s" can be calculated from the
size calculated in other file. The first integral "i"
esents fraction of power that is able to enter the ion
-2
0
2
4
6
8
10
12
14
16
8
6
4
2
-2

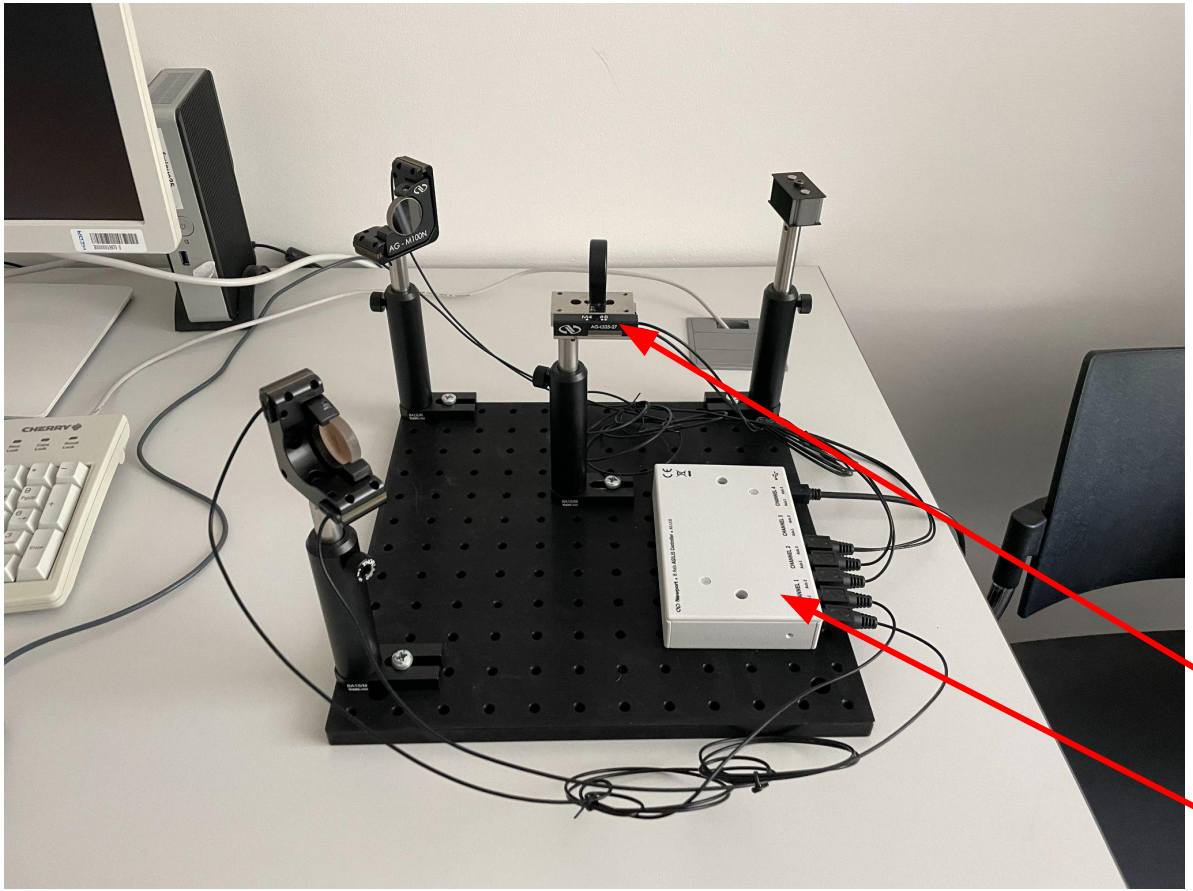


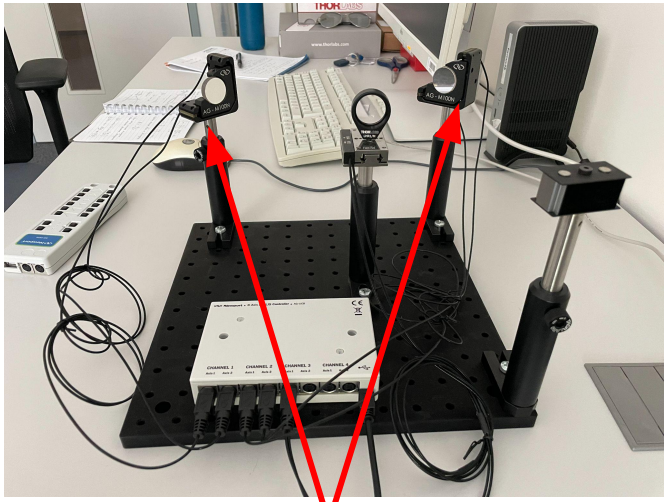




Successful focusing of Test laser through full setup distance!

Motorized Optical Element Controllers



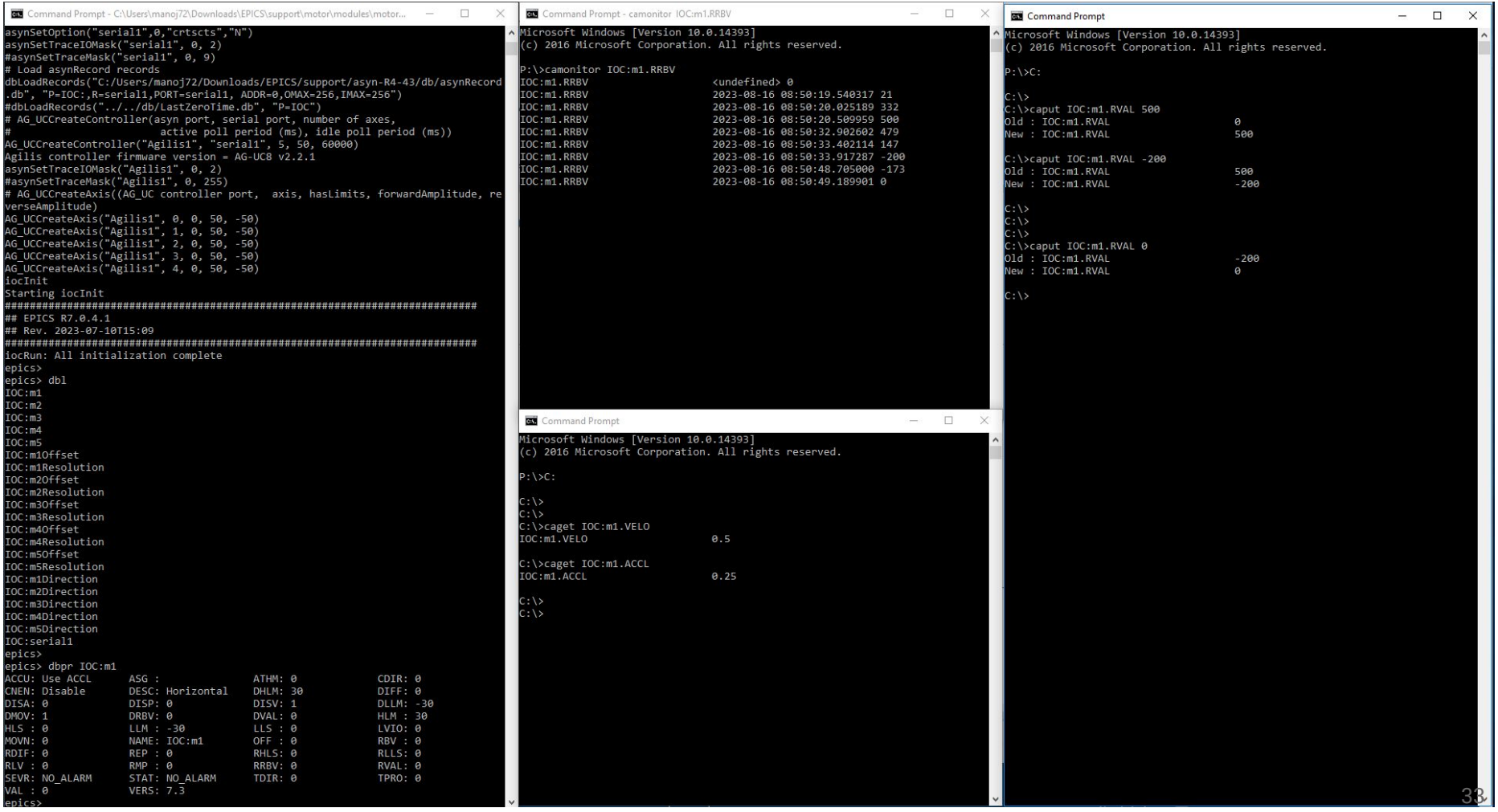


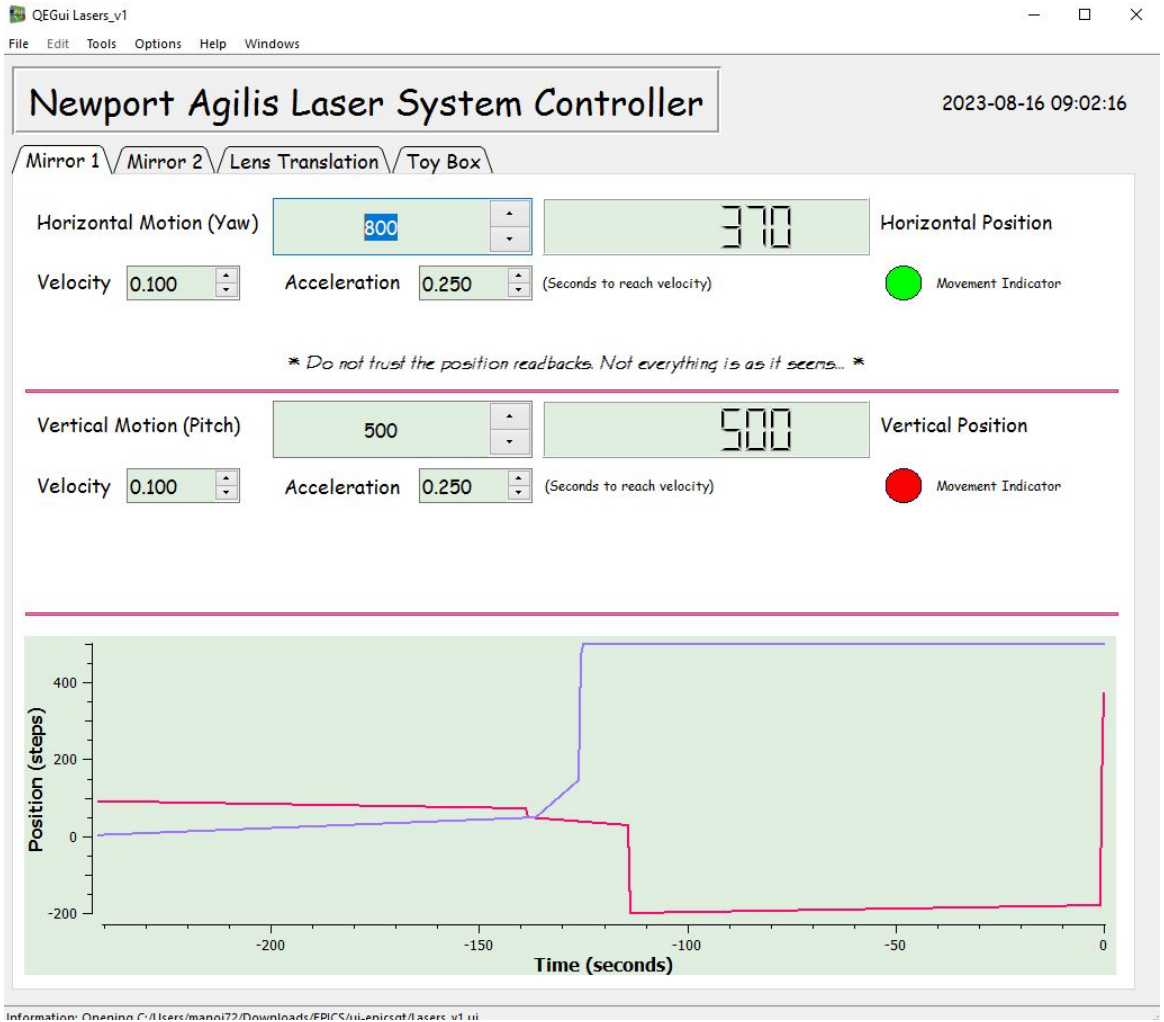
Newport Piezo Motorized Mirrors

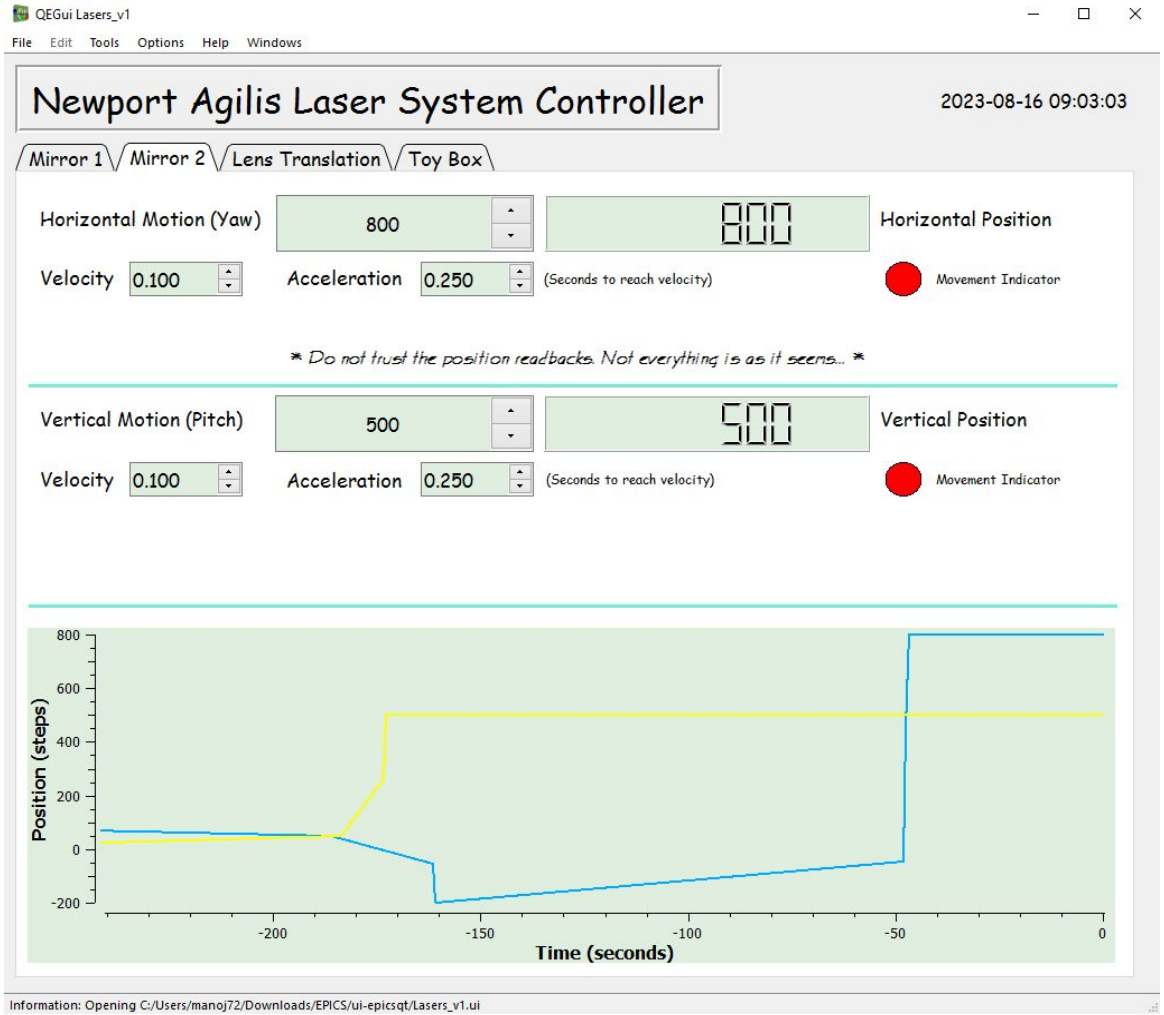
Newport Translation Stage with test Lens

Newport Agilis Motor Controller

Motorized Mirrors and Translation Stage for Keplerian Telescope Lens


Command Prompt - C:\Users\manoj72\Downloads\EPICS\support\motor\modules\motor...
asynSetOption("serial1",0,"crtscts","N")
asynSetTraceIOMask("serial1", 0, 2)
#asynSetTraceMask("serial1", 0, 9)
Load asynRecord records
dbLoadRecords("C:/Users/manoj72/Downloads/EPICS/support/asyn-R4-43/db/asynRecord.db", "P=IOC:,R=serial1,PORT=serial1, ADDR=0,OMAX=256,IMAX=256")
#dbLoadRecords("../../db/LastZeroTime.db", "P=IOC")
AG_UCCreateController(asyn port, serial port, number of axes,
active poll period (ms), idle poll period (ms))
AG_UCCreateController("Agilis1", "serial1", 5, 50, 60000)
Agilis controller firmware version = AG-UC8 v2.2.1
asynSetTraceIOMask("Agilis1", 0, 2)
#asynSetTraceMask("Agilis1", 0, 255)
AG_UCCreateAxis((AG_UC controller port, axis, hasLimits, forwardAmplitude, reverseAmplitude)
AG_UCCreateAxis("Agilis1", 0, 0, 50, -50)
AG_UCCreateAxis("Agilis1", 1, 0, 50, -50)
AG_UCCreateAxis("Agilis1", 2, 0, 50, -50)
AG_UCCreateAxis("Agilis1", 3, 0, 50, -50)
AG_UCCreateAxis("Agilis1", 4, 0, 50, -50)
iocInit
Starting iocInit
##
EPICS R7.0.4.1
Rev. 2023-07-10T15:09
##
iocRun: All initialization complete
epics>
epics> dbl
IOC:m1
IOC:m2
IOC:m3
IOC:m4
IOC:m5
IOC:m1Offset
IOC:m1Resolution
IOC:m2Offset
IOC:m2Resolution
IOC:m3Offset
IOC:m3Resolution
IOC:m4Offset
IOC:m4Resolution
IOC:m5Offset
IOC:m5Resolution
IOC:m1Direction
IOC:m2Direction
IOC:m3Direction
IOC:m4Direction
IOC:m5Direction
IOC:serial1
epics>
epics> dbpr IOC:m1
ACCU: Use ACCL ASG : ATHM: 0 CDIR: 0
CNEN: Disable DESC: Horizontal DHLM: 30 DIFF: 0
DISA: 0 DISP: 0 DISV: 1 DLLM: -30
DMOV: 1 DRBV: 0 DVAL: 0 HLM : 30
HLS : 0 LLM : -30 LLS : 0 LVIO: 0
MOVN: 0 NAME: IOC:m1 OFF : 0 RBV : 0
RDIF: 0 REP : 0 RHLS: 0 RLLS: 0
RLV : 0 RMP : 0 RRBV: 0 RVAL: 0
SEVR: NO_ALARM STAT: NO_ALARM TDIR: 0 TPRO: 0
VAL : 0 VERS: 7.3
epics>
Command Prompt - camonitor IOC:m1.RRBV
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
P:\>camonitor IOC:m1.RRBV
IOC:m1.RRBV <undefined> 0
IOC:m1.RRBV 2023-08-16 08:50:19.540317 21
IOC:m1.RRBV 2023-08-16 08:50:20.025189 332
IOC:m1.RRBV 2023-08-16 08:50:20.509959 500
IOC:m1.RRBV 2023-08-16 08:50:32.902602 479
IOC:m1.RRBV 2023-08-16 08:50:33.402114 147
IOC:m1.RRBV 2023-08-16 08:50:33.917287 -200
IOC:m1.RRBV 2023-08-16 08:50:48.705000 -173
IOC:m1.RRBV 2023-08-16 08:50:49.189901 0
Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
P:\>C:
C:\>
C:\>
C:\>caget IOC:m1.VELO
IOC:m1.VELO 0.5
C:\>caget IOC:m1.ACCL
IOC:m1.ACCL 0.25
C:\>
C:\>
Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
P:\>C:
C:\>
C:\>caput IOC:m1.RVAL 500
Old : IOC:m1.RVAL 0
New : IOC:m1.RVAL 500
C:\>caput IOC:m1.RVAL -200
Old : IOC:m1.RVAL 500
New : IOC:m1.RVAL -200
C:\>
C:\>
C:\>
C:\>caput IOC:m1.RVAL 0
Old : IOC:m1.RVAL -200
New : IOC:m1.RVAL 0
C:\>


QEGui Lasers_v1
File Edit Tools Options Help Windows
Newport Agilis Laser System Controller
2023-08-16 09:02:16
Mirror 1 Mirror 2 Lens Translation Toy Box
Horizontal Motion (Yaw)
800
370
Horizontal Position
Velocity 0.100
Acceleration 0.250
(Seconds to reach velocity)
Movement Indicator
* Do not trust the position readbacks. Not everything is as it seems... *
Vertical Motion (Pitch)
500
500
Vertical Position
Velocity 0.100
Acceleration 0.250
(Seconds to reach velocity)
Movement Indicator
Position (steps)
400
200
0
-200
-200
-150
-100
-50
0
Time (seconds)


QEGui Lasers_v1
File Edit Tools Options Help Windows
Newport Agilis Laser System Controller
2023-08-16 09:03:03
Mirror 1 Mirror 2 Lens Translation Toy Box
Horizontal Motion (Yaw)
800
800
Horizontal Position
Velocity 0.100
Acceleration 0.250
(Seconds to reach velocity)
Movement Indicator
* Do not trust the position readbacks. Not everything is as it seems... *
Vertical Motion (Pitch)
500
500
Vertical Position
Velocity 0.100
Acceleration 0.250
(Seconds to reach velocity)
Movement Indicator
Position (steps)
800
600
400
200
0
-200
-200
-150
-100
-50
0
Time (seconds)
Information: Opening C:/Users/manoj72/Downloads/EPICS/ui-epicsqt/Lasers_v1.ui

Thank You!

References

[1] M. Martschini, et al., Selective laser photodetachment of intense atomic and molecular negative ion beams with the ILIAS RFQ ion beam cooler, Int. J. Mass Spectrom. (2017), http://dx.doi.org/10.1016/j.ijms.2016.12.015.

[2] W. Kutschera, Applications of accelerator mass spectrometry, Int. J. Mass Spectr. 349–350 (2013) 203, http://dx.doi.org/10.1016/j.ijms.2013.05.023.

[3] A. Wieser, et al., "Detection of ^{135}CS & ^{137}CS in environmental samples by AMS", Nuclear Inst. and Methods in Physics Research B., vol. 538, 2023, pp. 36–40, https://doi.org/10.1016/j.nimb.2023.02.013.

[4] H.-A. Synal, Developments in accelerator mass spectrometry, Int. J. Mass Spectr. 349–350 (2013) 192, http://dx.doi.org/10.1016/j.ijms.2013.05.008.

[5] "Dreams." *ChETEC*, www.chetec-infra.eu/ta/tna-labs/hzdr-dreams/.

[6] A. Wieser, What is Alex doing all the time?, 7 June 2023. Powerpoint Presentation

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[7] “Dating.” *The Smithsonian Institution’s Human Origins Program*, 16 Aug. 2022, https://humanorigins.si.edu/evidence/dating.

[8] C. Marek (2018), AMS measurements of ^{36}Cl using isobar suppression via laser photodetachment, Masters Thesis, University of Vienna

[9] M. Martschini (2012), Development of methods for isobar suppression in AMS and measurements of 36Cl with the VERA 3-MV-tandem accelerator, PhD thesis, University of Vienna.

[10] Z. Sóti, et al. “Karlsruhe Nuclide Chart – new 10th edition 2018 .” *EPJ Nuclear Sciences & Technologies*, vol. 5, 2019, p. 6, https://doi.org/10.1051/epjn/2019004.

[11] “The Beginner’s Guide on Spot Size of Laser Beam.” *Gentec*, www.gentec-eo.com/blog/spot-size-of-laser-beam.

[12] “Beam Expander Design Comparison: Keplerian and Galilean.” *Thorlabs, Inc.*, www.thorlabs.us/newgrouppage9.cfm?objectgroup_id=14648.

References

[13] Compact Laser Modules with Phono Jack, *Thorlabs*, Inc., www.thorlabs.de/newgrouppage9.cfm?objectgroup_id=1487&pn=CPS532.

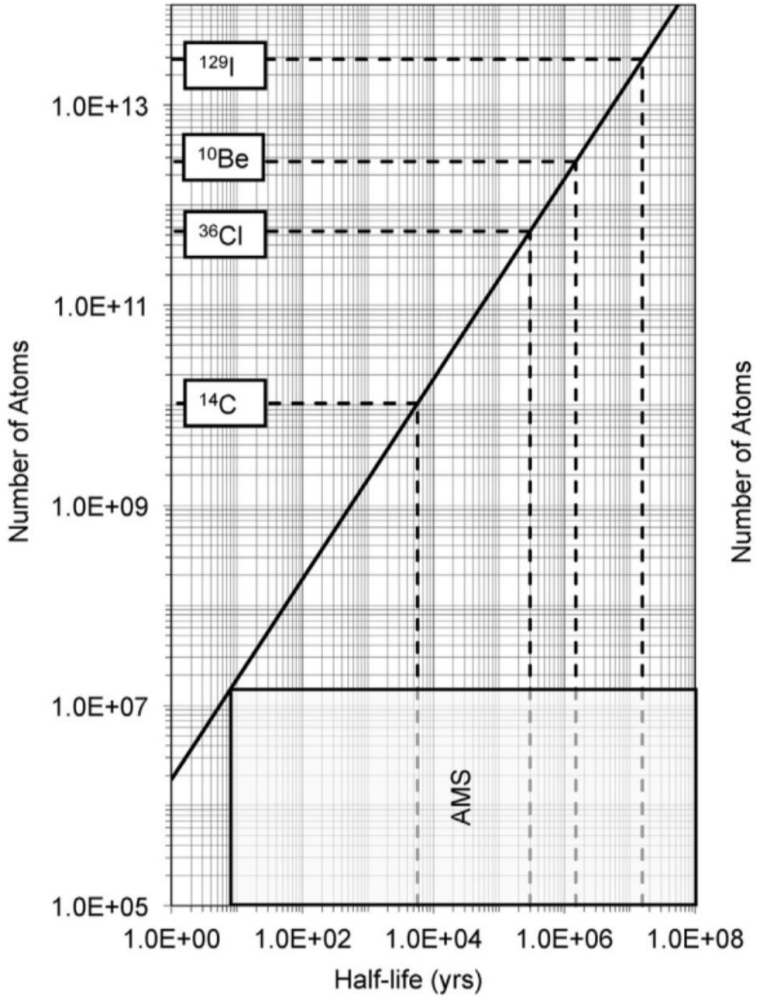
[14] “Verdi - High Performance CW Green Lasers.” *Coherent*, www.coherent.com/lasers/cw-solid-state/verdi.

Extra Slides

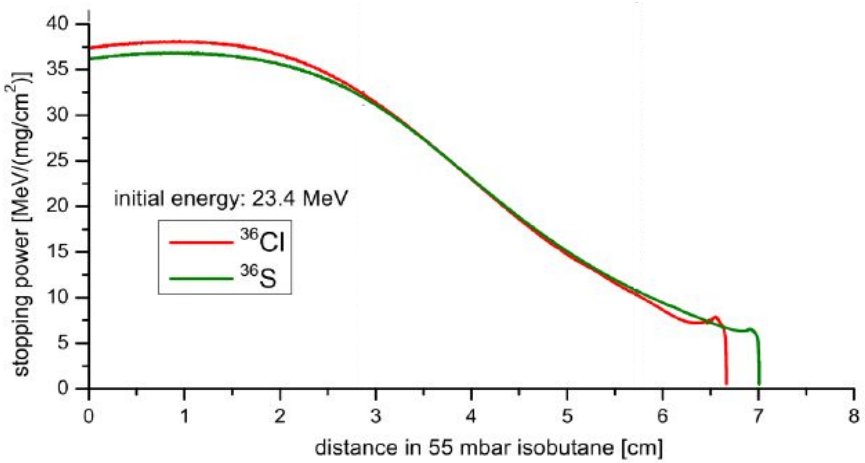
Long half lives make decay radiation detection difficult

~ Comparison decay counting versus AMS for the amounts of nuclides needed to perform a measurement with 1% precision based on counting statistics.

(Synal, Hans-Arno. “Developments in accelerator mass spectrometry.”)



Using Stopping Power to Separate Isobars



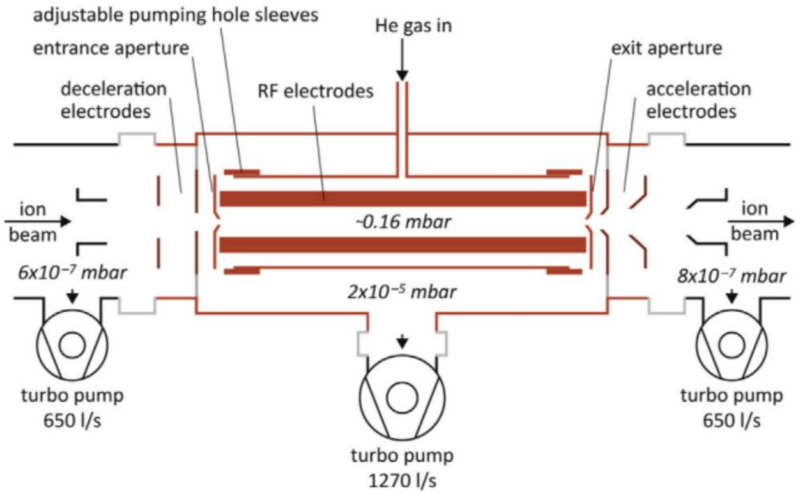
~ The total stopping power of ^{36}Cl and ^{36}S with initial energy of 23.4MeV is plotted against the travel distance [8] [9]

- Classical AMS uses difference in energy loss characteristics of the ions when passing through matter [1]
  - Depends on relative difference in proton number ($\Delta Z/Z$) and beam energy

Ion Cooler

- Ion beam must be decelerated to maximize interaction time with the laser
- Helium buffer gas to cool the ion beam
- Quadrupole field used to confine ions

- Entrance and Exit apertures are 3mm in diameter
  - We must optimize laser to fit through the apertures and maximize power deposited in the ion cooler



~ Schematic Drawing of Ion Cooler at Vienna
(M. Martschini, et al., Selective laser photodetachment of intense atomic and molecular negative ion beams with the ILIAS RFQ ion beam cooler, Int. J. Mass Spectrom. (2017), http://dx.doi.org/10.1016/j.ijms.2016.12.015)